

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 711 Applied Cryptography — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Aumasson — Serious Cryptography, 2nd edition	The primary text, and the right level for this course. Written by a practising cryptographer for engineers who must use the primitives correctly, and it is consistently honest about what is and is not settled. Library copy.
Boneh & Shoup — A Graduate Course in Applied Cryptography (free PDF)	Free in full, and the rigorous companion. Where this course states a property informally, Boneh and Shoup give the definition and the proof. Modules 03 through 07 follow its structure.
Katz & Lindell — Introduction to Modern Cryptography	The standard text on the definitional approach — why security is defined as a game against an adversary, which is Module 01 §3 and is the idea that makes the rest rigorous. Library copy.
The libsodium documentation (free)	The course's reference implementation, and the embodiment of Module 01's rule. A small, opinionated, misuse-resistant API — read it as a design argument rather than only as documentation.
Cryptographic Right Answers, current edition (free)	A short list of what to use, maintained by practitioners who review real systems. Read it after Module 04 and again after Module 11 — it will read differently.
RFC 8446 (TLS 1.3) and RFC 9106 (Argon2) (free)	Module 08's and Module 09's primary sources. TLS 1.3 is worth reading as an example of a protocol deliberately simplified to remove misuse, which is this course's thesis in standards form.

Module-by-module map

Module 01 — What Cryptography Provides

Resource	What it covers for this module
Aumasson — Serious Cryptography, chapters 1 and 2	§1 and §3 at this course's level , and the treatment of what security definitions are for is the clearest available.
Boneh & Shoup — A Graduate Course in Applied Cryptography, chapter 2 (free)	§3 rigorously — the definitions, the advantage formulation, and what a security proof actually establishes.
Cryptographic Right Answers (free)	§4 as a practical list , maintained by people who review real systems for a living. Short, and worth rereading after Module 11.
The libsodium documentation (free)	§4's rule embodied in an API — read what it refuses to let you do, which is the design argument.

Module 02 — Randomness

Resource	What it covers for this module
Aumasson — Serious Cryptography, chapter 2	The whole module , with more detail on the entropy pool design and the hardware source question.
Heninger et al. — Mining Your Ps and Qs (free)	§3's third failure, measured at internet scale — the study that found hundreds of thousands of devices with shared or factorable keys.
Myths about /dev/urandom (free)	§2's folklore, dismantled carefully — read it once and stop worrying about this.
NIST SP 800-90A/B/C (free)	The standards for CSPRNG construction and entropy assessment — reference rather than reading, but worth knowing exists.

Module 03 — Symmetric Encryption

Resource	What it covers for this module
Boneh & Shoup, chapters 3 through 5 (free)	§1 through §3 rigorously — including the proofs that CTR and CBC achieve the Module 01 §3 definition given their requirements.
Aumasson — Serious Cryptography, chapters 4 and 5	The modes and stream ciphers at engineering level , with the practical hazards foregrounded.
RFC 8439 — ChaCha20 and Poly1305 (free)	§3's specification , and short enough to implement from — which is one of Project 1's options.
The ECB penguin, and why it persists (free)	§2's demonstration, with the nuance about what the image does and does not prove.

Module 04 — Authenticated Encryption

Resource	What it covers for this module
Boneh & Shoup, chapters 6 through 9 (free)	MACs and authenticated encryption with the proofs — including why encrypt-then-MAC is generically secure and the others are not.
Aumasson — Serious Cryptography, chapters 7 through 9	§1 through §3 at engineering level , with the construction comparisons laid out practically.
RFC 5116 and RFC 8452 (AEAD interface; AES-GCM-SIV) (free)	§3's interface as standardised , and the misuse-resistant construction's specification and rationale.
Rizzo & Duong on padding oracles (free)	§4's argument, demonstrated — plaintext recovery from nothing but distinguishable error behaviour.

Module 05 — Hash Functions

Resource	What it covers for this module
Aumasson — Serious Cryptography, chapters 6 and 7	The whole module , with the construction internals and the practical choice guidance.
Boneh & Shoup, chapters 8 and 14 (free)	§1's definitions and §3's HMAC proof — including why a MAC does not require collision resistance.
The SHAttered SHA-1 collision (free)	§1's demonstration , with the colliding PDF pair and the cost analysis.
Certificate Transparency (free)	§4's Merkle trees in production — and the infrastructure Module 07 §4 depends on.

Module 06 — Key Exchange and Public Keys

Resource	What it covers for this module
Boneh & Shoup, chapters 10 through 12 (free)	§1 through §3 rigorously — the assumptions, their relationships, and what the reductions establish.
Aumasson — Serious Cryptography, chapters 10 through 12	RSA, Diffie–Hellman, and elliptic curves at engineering level , with the practical hazards foregrounded.
Bernstein — Curve25519, and the SafeCurves criteria (free)	§4's design argument — what makes a curve hard to misuse, stated as explicit criteria.
Diffie & Hellman — New Directions in Cryptography (free)	§1 in the original, from 1976 — short, readable, and one of the genuinely pivotal papers in computing.

Module 07 — Signatures and Certificates

Resource	What it covers for this module
Aumasson — Serious Cryptography, chapter 12	§1's schemes , with the ECDSA nonce hazard explained properly.
Ristic — Bulletproof TLS and PKI	§2 through §4 as the practical reference — the best single treatment of the certificate ecosystem. Library copy.
Certificate Transparency (free)	§3's mitigation , and worth reading as a design response to an unfixable trust model.
RFC 6979 — Deterministic ECDSA (free)	§1's fix for the sharpest edge — and a good example of removing a failure mode rather than warning about it.

Module 08 — TLS, as the Worked Example

Resource	What it covers for this module
RFC 8446 — TLS 1.3 (free)	§1 and §2 in the original — and the handshake section is genuinely readable, which is unusual for an RFC.
Ristic — Bulletproof TLS and PKI	The whole module as the practical reference , including the configuration guidance for §4. Library copy.
The Illustrated TLS 1.3 Connection (free)	§1 byte by byte , annotated — the single best way to see the handshake concretely.
SSL Labs server test, and testssl.sh (free)	§4's checks, automated — run against your own services, and read the explanations rather than only the grade.

Module 09 — Passwords and Key Derivation

Resource	What it covers for this module
RFC 9106 — Argon2 (free)	§2's recommendation in the original , including the parameter selection guidance and the variant rationale.
OWASP — Password Storage Cheat Sheet (free)	§2 and §4 as current practice , maintained and updated as hardware moves — which matters for the parameters.
RFC 5869 — HKDF (free)	§3's construction , short, with the extract-then-expand rationale clearly stated.
NIST SP 800-63B, the memorized secret section (free)	§4's evidence-based guidance — and notable as a standard that reversed long-standing advice on the basis of measurement.

Module 10 — Key Management

Resource	What it covers for this module
Google — Building Secure and Reliable Systems, the key management chapters (free PDF)	§1 through §3 from operators — and free in full. The strongest practical treatment available.
AWS KMS and Google Cloud KMS design documentation (free)	§2's pattern as implemented at scale — read the envelope encryption and key hierarchy sections regardless of which platform you use.
Google Tink (free)	§2 and §3 as a library — keyset rotation and key identifiers are built in, which is the design argument made concrete.
NIST SP 800-57 — Key Management Recommendations (free)	§1's lifecycle, exhaustively — reference rather than reading, and useful for the stages people skip.

Module 11 — Implementation Hazards

Resource	What it covers for this module
Aumasson — Serious Cryptography, the implementation chapters	§1 and §2 at engineering level , with the constant-time patterns and the misuse consequences.
BearSSL — Constant-Time Cryptography (free)	§1's best single reference — thorough, practical, and honest about what the compiler does to you.
Bernstein — Cache-timing attacks on AES (free)	§1's threat demonstrated — and the paper that changed how AES is implemented in software.
Joux — Authentication failures in NIST GCM (free)	§2's GCM row, in the original — why nonce reuse costs the authentication key specifically.

Module 12 — Post-Quantum Cryptography

Resource	What it covers for this module
NIST Post-Quantum Cryptography project (free)	§3's standards and the selection rationale — free, and considerably more measured than most commentary on the subject.
NSA CNSA 2.0 and the NIST migration guidance (free)	§2's timelines as published by parties who must plan them — useful as a datum rather than a prediction.
Bernstein & Lange — Post-quantum cryptography (free)	§1 and §3 from researchers in the field , including candid assessment of the assumptions.
Open Quantum Safe (free)	§3 and §4 as working code — hybrid TLS you can actually deploy and measure.

Module 13 — Claiming Cryptographic Security Honestly

Resource	What it covers for this module
Boneh & Shoup, the introduction and chapter 2 (free)	§3's account of what a proof establishes , from the people who write them — and notably careful about the limits.
Koblitz & Menezes — Another Look at “Provable Security” (free)	§3's caveats argued forcefully — a useful corrective, and contested, which is itself instructive.
Green — A Few Thoughts on Cryptographic Engineering (free)	§1 and §2 in practice — consistently good at separating what a system claims from what it provides.
Anderson — Security Engineering, the cryptography chapter (free PDF)	§4's placement of this subject within security as a whole , and consistent with this course's ordering.